BEING ETALE IS ETALE LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof line 7173. Reduces the claim to four clauses.
- Zariski locality on source and target and preservation under precomposition for etale morphisms. Morphisms, Lemmas morphisms-lemma-etale-characterize and morphisms-lemma-composition-etale, invoked at proof lines 7174-7176. Supply clauses (1)-(3).
- Permanence of etaleness under an etale surjection on the source. Same chapter, Lemma lemma-syntomic-smooth-etale-permanence, invoked at proof line 7177. Supplies clause (4).

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KM: sources/stacks-descent/pinned-original-source.tex, source lines 4640–4640. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GN: sources/stacks-chow/tags.txt, source lines 3234–3234. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GJ: sources/stacks-chow/tags.txt, source lines 3230–3230. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *The property $\mathcal{P}(f) =$"f is étale" is étale local on the source.*

Proof. Combine Lemma 036H with Morphisms, Lemma 02GJ (local for Zariski on source and target), Morphisms, Lemma 02GN (pre-composing), and Lemma 02KM (part (4)). □